

# Week 6 Problem Set

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1. Start a new document, problemset6.Rmd, in your soc10problemsets repository. (You can download a template here, but be sure to save it in your own soc10problemsets repo). Create a network based on 5 people in our class! (You can use the “class\_activities.csv” file from Canvas if helpful or use the code in the template). Using **igraph**, create a network diagram based on who has worked with who in our class activities this quarter (you may want to use **graph\_from\_data\_frame()** if you read in the csv). Then plot the network. (Note: a network with only 5 people, excluding ties to everyone outside of these 5, is sufficient). **Commit** your changes.

[Note: the scoring guidelines in brackets can be deleted before submitting. Create a network with igraph (1 point), and plot it (0.5 points).]

```
library(readr)
library(RCurl)
```

```
## Warning: package 'RCurl' was built under R version 4.5.2
```

```
library(magrittr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(igraph)
```

```
##
## Attaching package: 'igraph'
```

```
## The following objects are masked from 'package:dplyr':
##
##   as_data_frame, groups, union
```

```

## The following objects are masked from 'package:stats':
##
##   decompose, spectrum

## The following object is masked from 'package:base':
##
##   union

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/class_activities.csv")

# download class activities data (from github)
class_activities <- read_csv(url)

## Rows: 71 Columns: 3

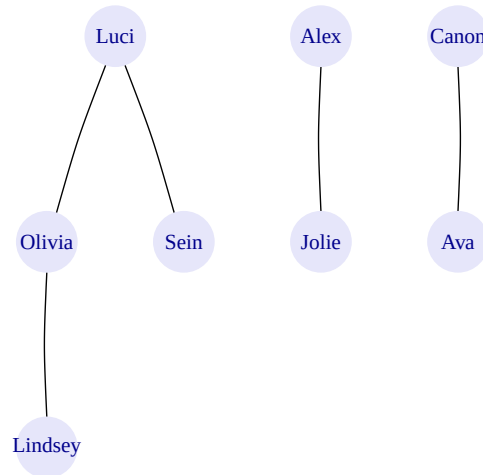
## -- Column specification -----
## Delimiter: ","
## chr (2): name_1, name_2
## dbl (1): n
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

# Network using 5 people
el <- data.frame(
  person_1 = c("Lindsey", "Olivia", "Jolie", "Luci", "Canon"),
  person_2 = c("Olivia", "Luci", "Alex", "Sein", "Ava")
) %>%
  as.matrix()

# create network
net <- graph_from_edgelist(el, directed = FALSE)

# now I will plot the network
plot(net, vertex.size = 30,
      vertex.color = "lavender",
      vertex.frame.color = NA,
      vertex.label.cex = .7,
      edge.curved = .05,
      edge.arrow.size = .3,
      edge.width = .7,
      edge.color = "gray1")

```



2. Choose a centrality measure, and explain how you would interpret this in the context of the class activities data. Who is/are the most central actor(s) according to this measure? **Commit** your changes.

I chose degree centrality because it is used to measure how many connections each student has in the network for class collaboration like shown in class. Hence, in this context, a higher degree centrality means that a student has more collaboration ties within this network of classmates (has worked with more people in this selected group). For example, it shows that Olivia and Luci are the most central actors as they each have the highest number of direct collaboration ties in the network (2).

```
# Calculate degree centrality
degree_centrality <- degree(net)
```

```
# View degree centrality scores
degree_centrality
```

```
## Lindsey  Olivia    Luci   Jolie   Alex   Sein   Canon   Ava
##         1         2         2         1         1         1         1
```

```
# Identify most central actor(s)
most_central <- names(degree_centrality[
  degree_centrality == max(degree_centrality)
])
```

```
most_central
```

```
## [1] "Olivia" "Luci"
```

[Correctly use a centrality measure (0.5 points) and interpret its meaning (0.5 points). Identify the most central actor(s) (0.5 points).]

3. Now download the IPCC co-authorship data (you can use the code included in the template). Using the IPCC co-authorship data, create a network graph of all co-authors who share at least 2 publications together, excluding those with no network ties, similar to what we do in section @ref(co-authorship-patterns). Graph the data using vector scores, rather than betweenness scores, to plot the vertex sizes. Does this make a difference in our interpretation of the network? **Commit** your changes.

[Create a network graph of all co-authors with correct number of ties (1 point). Use vector scores rather than betweenness scores to scale vertices (0.5 points). Describe the findings and potential implications (1 point).]

Eigenvector centrality changes the interpretation of the network compared to betweenness centrality. Betweenness centrality identifies authors who act as bridges between different parts of the network, as eigenvector centrality identifies authors who are highly connected to other well connected/influential authors. Therefore, eigenvector centrality emphasizes the core influential group in the IPCC co-authorship network instead of brokerage positions. Hence, the implication of this is that influence in this network is focused within a core group of authors instead of spread through intermediaries connected to various parts of the network.

```
library(readr)
library(RCurl)
library(magrittr)
library(dplyr)

url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_co_authorship.csv")
ipcc_authors <- read_csv(url, skip = 1)

ipcc_authors %<>%
mutate_all(~replace(., is.na(.), 0))

ipcc_authors %<>%
select(order(colnames(.)))

ipcc_authors %<>%
select(-Author) %>%
  as.matrix()

# replace 1s with 0s
ipcc_authors[ipcc_authors == 1] = 0

# replace NAs with 0s
ipcc_authors[is.na(ipcc_authors)] = 0

# vector of authors who do not collaborate with others
no_colab <- which(apply(ipcc_authors, 1, sum, na.rm = TRUE) == 0)
which(apply(ipcc_authors, 2, sum, na.rm = TRUE) == 0)
```

|    |            |               |         |       |
|----|------------|---------------|---------|-------|
| ## | Abdel-Aziz | Acosta-Moreno | Ahammad | Ahmed |
| ## | 1          | 2             | 5       | 6     |
| ## | Akimoto    | Attzs         | Barua   | Bigio |

|    |                 |                    |                 |                    |
|----|-----------------|--------------------|-----------------|--------------------|
| ## | 8               | 10                 | 14              | 17                 |
| ## | Blanco_Gabriel  | Blanco_Hilda       | Bouille         | Brewer             |
| ## | 18              | 19                 | 23              | 24                 |
| ## | Broome          | Bruvoll            | Cabeza          | Chen_Wenying       |
| ## | 25              | 28                 | 30              | 35                 |
| ## | Chimanikire     | Chingambo          | Clark           | Clift              |
| ## | 37              | 38                 | 40              | 42                 |
| ## | Conte-Grand     | Cooke              | Corbera         | Cottier            |
| ## | 43              | 44                 | 45              | 46                 |
| ## | Creutzig        | Cruz-Nunez         | de Coninck      | De-La-Vega-Navarro |
| ## | 47              | 48                 | 51              | 52                 |
| ## | Delgado-Ramos   | Demkine            | Dewar           | Dimitriu           |
| ## | 54              | 55                 | 57              | 60                 |
| ## | Dubash          | El-Gizouli         | Elsiddig        | Essandoh-Yeddu     |
| ## | 62              | 66                 | 67              | 68                 |
| ## | Fifita          | Finus              | Fleurbay        | Frankel            |
| ## | 71              | 73                 | 76              | 77                 |
| ## | Fullerton       | Fulton             | Fungtammasan    | Garrido-Vazquez    |
| ## | 78              | 79                 | 80              | 84                 |
| ## | Gomez-Echeverri | Gutierrez-Espeleta | Harvey          | Hassan             |
| ## | 89              | 96                 | 104             | 105                |
| ## | Honnery         | Huang_Luxin        | Huang_Yongfu    | Inaba              |
| ## | 109             | 111                | 112             | 113                |
| ## | Infield         | Jaenicke           | Jafari          | Karrouk            |
| ## | 114             | 116                | 117             | 126                |
| ## | Kartha          | Khennas            | Kim_Suduk       | Kim_Yong-Gun       |
| ## | 127             | 128                | 130             | 131                |
| ## | Klasen          | Kolstad            | Labandeira      | Lee                |
| ## | 132             | 134                | 141             | 144                |
| ## | Lefevre         | Ling               | Liphoto         | Llanes-Regueiro    |
| ## | 145             | 147                | 149             | 150                |
| ## | Loeschel        | Lwasa              | Machado-Filho   | Mathur             |
| ## | 151             | 154                | 155             | 159                |
| ## | McKinnon        | Messouli           | Meyer           | Montero            |
| ## | 161             | 163                | 164             | 168                |
| ## | Morgan          | Mulugetta          | Mundaca         | Murakami_Shuzo     |
| ## | 169             | 171                | 172             | 174                |
| ## | Muvundika       | Nenov              | Newman          | Nimir              |
| ## | 175             | 180                | 181             | 182                |
| ## | Nzunda          | Ofosu-Ahenkorah    | Okereke         | Olivier            |
| ## | 184             | 185                | 186             | 187                |
| ## | Ouyang          | Pan                | Parikh_Jyoti    | Parikh_Kirit       |
| ## | 188             | 191                | 192             | 193                |
| ## | Paterson        | Perczyk            | Perez-Arriaga   | Pyke               |
| ## | 194             | 197                | 198             | 202                |
| ## | Ramakrishna     | Rauscher           | Robledo-Abad    | Romanovskaya       |
| ## | 203             | 205                | 211             | 213                |
| ## | Sagar           | Santalla           | Seroa-da-Motta  | Sheikho            |
| ## | 217             | 218                | 222             | 224                |
| ## | Shittu          | Somanathan         | Sperling_Daniel | Sperling_Frank     |
| ## | 226             | 232                | 233             | 234                |
| ## | Suliman         | Teng               | Tiwari          | Torres-Martinez    |
| ## | 241             | 244                | 245             | 246                |
| ## | Uddin           | Venables           | Vilarino        | Wen                |

|    |           |              |            |        |
|----|-----------|--------------|------------|--------|
| ## | 249       | 254          | 257        | 259    |
| ## | Wiener    | Yetano-Roche | Zain-Ahmed | Zatari |
| ## | 261       | 266          | 267        | 268    |
| ## | Zhou_Dadi | Zhou_Peter   | Zou        | Zylicz |
| ## | 270       | 271          | 272        | 273    |

```
ipcc_authors <- ipcc_authors[-no_colab, -no_colab]
```

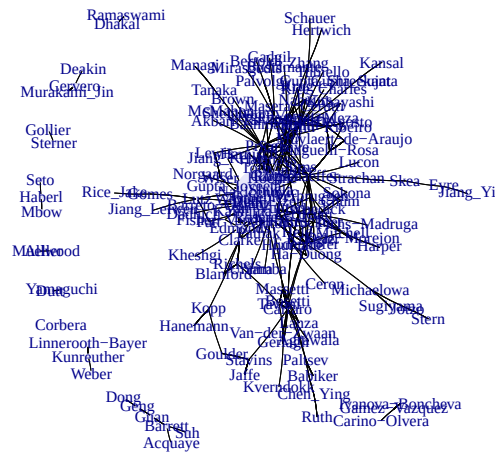
```
ipcc_net <- graph_from_adjacency_matrix(ipcc_authors, mode = "undirected")
```

```
## Warning: The 'adjmatrix' argument of 'graph_from_adjacency_matrix()' must be symmetric
## with mode = "undirected" as of igraph 1.6.0.
## i Use mode = "max" to achieve the original behavior.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
V(ipcc_net)$vector <- evcent(ipcc_net)$vector
```

```
## Warning: 'evcent()' was deprecated in igraph 2.0.0.
## i Please use 'eigen_centrality()' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
plot(ipcc_net,
      vertex.size = V(ipcc_net)$vector/max(V(ipcc_net)$vector) * 20,
      vertex.color = "lavender",
      vertex.frame.color = NA,
      vertex.label.cex = .5,
      edge.curved = .05,
      edge.width = .1,
      edge.color = "gray1")
```



4. Let's say you want to give one of the IPCC authors a memo to disseminate to the others. Who would you choose? Which measure of centrality would help you arrive at your answer? **Commit** your changes.

[Identify a reasonable strategy for dissemination (1 point). Discuss a reasonable centrality measure that aligns with strategy (0.5 points). Correctly identify the node with highest value for said centrality measure (0.5 points).]

In order to disseminate a memo efficiently through the IPCC co-authorship network I would choose an author who is centrally located in the most connected part of the network, to allow the information to spread quickly to others. Hence, eigenvector centrality is the best measure for this as it identifies authors who are not just well connected themselves but also connected to other highly influential and connected authors. Therefore, this permits it to be easier to find the influence in a collaborative diffusion process. Because of the eigenvector centrality rankings I would then choose the author with the highest eigenvector centrality score which happens to be Uerge-Vorsatz as the starting point for the dissemination because it is the most influential node in the network.

```
# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_co_authorship.csv")

# download authorship data (from github)
ipcc_authors <- read_csv(url, skip = 1)

# replace NA's with 0's
ipcc_authors %<>%
mutate_all(~replace(., is.na(.), 0))
```

```

# Save author names
author_names <- ipcc_authors[[1]]

# Create matrix
ipcc_matrix <- as.matrix(ipcc_authors[, -1])

rownames(ipcc_matrix) <- author_names
colnames(ipcc_matrix) <- author_names

# Create network
ipcc_network <- graph_from_adjacency_matrix(ipcc_matrix,
mode = "upper",
weighted = TRUE,
diag = FALSE)

# Keep only authors with at least 2 shared publications
ipcc_network_2 <- delete_edges(ipcc_network,
E(ipcc_network)[weight < 2])

# Remove isolated authors
ipcc_network_2 <- delete_vertices(ipcc_network_2,
degree(ipcc_network_2) == 0)

# Calculate degree centrality
degree_ipcc <- degree(ipcc_network_2)

# Find most central author
memo_author <- names(degree_ipcc[degree_ipcc == max(degree_ipcc)])
memo_author

```

```
## [1] "Uerge-Vorsatz"
```

5. Finally, plot the IPCC co-authorship data, but change the color/shape of the vertices so that these correspond to authors' gender. Describe what you see. What conclusions can you make about authorship and gender? (Note that you will need the IPCC\_cv.csv dataset. The code in the template has combined these datasets). **Commit** your changes.

[Correctly define a gender variable, and include an accurate label (1 point). Plot the IPCC co-authorship data with modified color/shape based on gender (0.5 points). Accurately describe the plot (0.5 points). Assess conclusions that can (or can't) be drawn (0.5 points).]

The gender variable is defined using the IPCC CV data set and recoded so the authors are grouped by gender such as male, female and other/unknown. Hence, this variable is added to the network vertices so every author has a gender attribute within the graph. When looking at the network it is plotted with betweenness centrality to scale the node size so more central authors appear larger. Additionally, vertex color is used to represent gender to compare the differences in representation for each gender within the network. Therefore, this plot shows that male authors are a larger portion of the network and are more central located which means highly connected. On the other hand, female authors appear less and are more on the outside (exterior) of the network. Overall, this draws the conclusion that there is an imbalance in the representation among genders and centrality in the IPCC co-authorship network, yet the visualization is not the cause of this difference so you cannot make a causal conclusion just that there is a difference in representation of gender and structure of the network.



```

library(readr)
library(RCurl)
library(magrittr)
library(dplyr)
library(igraph)

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_co_authorship.csv")

# download authorship data (from github)
ipcc_authors <- read_csv(url, skip = 1)

# replace NA's with 0's
ipcc_authors %<>%
  mutate_all(~replace(., is.na(.), 0))

# reorder column names
ipcc_authors %<>%
  select(order(colnames(.)))

# convert to matrix
ipcc_authors %<>%
  select(-Author) %>%
  as.matrix()

# replace 1s with 0s
ipcc_authors[ipcc_authors == 1] = 0

# replace NAs with 0s
ipcc_authors[is.na(ipcc_authors)] = 0

# vector of authors who do not collaborate with others
no_colab <- which(apply(ipcc_authors, 1, sum, na.rm = TRUE) == 0)

# then we can remove these from ipcc_authors
ipcc_authors <- ipcc_authors[-no_colab,-no_colab]

# create network
ipcc_net <- graph_from_adjacency_matrix(ipcc_authors,
                                       mode = "undirected")

# add vector centrality
V(ipcc_net)$betweenness <- betweenness(ipcc_net)

# input url from github
url <- getURL("https://raw.githubusercontent.com/tylermcdaniel/css/main/Data/IPCC_cv.csv")

# download cv data
ipcc_cvs <- read_csv(url)

# remove authors with no collaborations
ipcc_cvs %<>%
  slice(-no_colab)

```

```

#define gender
ipcc_cvs %<>%
  mutate(gender = case_match(Gender,
                             1 ~ "tomato",
                             2 ~ "steelblue",
                             .default = "gray"))

# add gender variable to vertices
V(ipcc_net)$gender <- ipcc_cvs$gender

# plot network
plot(ipcc_net,
     edge.width = .1,
     vertex.size = V(ipcc_net)$betweenness/max(V(ipcc_net)$betweenness) * 20,
     vertex.color = V(ipcc_net)$gender,
     vertex.alpha = 0.5,
     vertex.frame.color = NA,
     vertex.label.cex = .5,
     edge.curved = .05,
     edge.width = .1,
     edge.color = "gray1")

# Now add legend
legend("bottomright",
      legend = c("Male", "Female", "Other/Unknown"),
      pt.bg = c("tomato", "steelblue", "gray"),
      pch = 21,
      cex = 1,
      title = "Gender")

```

